Ibtida Ara

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Professional Summary

Final-year Computer Science and Engineering student with a strong academic background and practical experience in software development, problem-solving, and teamwork. Passionate about learning new technologies and building efficient, user-focused applications. Seeking an internship or entry-level software engineering position to apply technical skills while contributing to organizational success.

Skills

Languages: C, C++, Java, Python, JavaScript, PHP
Web & Database: HTML, CSS, JavaScript, PHP, MySQL
AI & Research: Python, Ollama, Google Colab, Jupyter Notebook
Design & UX: Figma, Miro, Wireframing, Prototyping, Heuristic Evaluation, Usability Testing
Embedded: Arduino, ESP32
Tools & Platforms: VS Code, IntelliJ IDEA, Jira, Microsoft Office Suite, Google Workspace, Overleaf
Version Control: Git, GitHub
Networking: Understanding of routing, switching, and network protocols (TCP/IP, DNS, DHCP)
OS & Systems: Linux (Ubuntu), Windows

Work Experience

United International University 13 Jul 2026 – Present
Undergraduate Teaching Assistant

CodeAlpha | Remote Aug 2026 – Present
ML Engineering Intern

- Selected as an ML Engineering Intern at CodeAlpha's remote internship program.
- Working on assigned machine learning tasks — data preprocessing, model training/evaluation, and applying ML workflows — to gain practical industry experience.
- Collaborating with the team on task deliverables and following structured internship guidelines.

Thiranex | Remote Jun 2026 – Aug 2026
UI/UX Design Intern

- Selected as a UI/UX Design Intern at Thiranex's internship program.
- Learned and applied UI/UX design principles, user research, and design workflows.
- Worked on assigned internship tasks to gain practical industry experience.

Education

United International University **CGPA: 3.91/4.00**
B.Sc. in Computer Science and Engineering (Session: 2023–2027)
Relevant Coursework: Object Oriented Programming, Data Structures & Algorithms I & II, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Data Mining, Machine Learning, Project Management

Dhaka City College **GPA: 5.00**
Higher Secondary Certificate (HSC), Science Group — Session: 2019–2021, Batch: 2021

Monipur High School **GPA: 5.00**
Secondary School Certificate (SSC), Science Group — Session: 2017–2019, Batch: 2019

Project Work

- **Green LLM Token Research** *(Python, Ollama, Google Colab)*: Studied whether prompt optimization can cut token use in student tasks without hurting answer quality, across 100 tasks, 4 small models, and 800 primary runs; early results show 23.67% fewer tokens on Gemma 3 4B and 17.16% on Ministral 3 3B. [GitHub]

- **Bangla Traffic Law MCQ Dataset & Benchmark** *(Legal AI, Dataset Design)*: Built a native Bangla benchmark of 299 hand-verified MCQs from the Bangladesh Road Transport Act, 2018, scored on accuracy, factual error, and hallucination to test how well language models know a local statute. [GitHub]
- **Aiva: From Friend to Partner** *(Figma, Miro, Usability Testing)*: HCI project on how an AI companion should handle a shift from friendly to romantic interaction; built a 24-screen high-fidelity prototype and tested it with 8 participants (85.4% task completion) after a 17-person survey and 5 interviews. [GitHub]
- **AI-Powered Hexapod for Autonomous & Wireless-Controlled Navigation** *(ESP32, Arduino, Bluetooth)*: Built a low-cost six-legged robot with 12 servos and a tripod gait, driven from a smartphone over Bluetooth, with ultrasonic obstacle detection that stops the robot near obstacles; designed for autonomous navigation and IMU-based self-balancing. [GitHub]
- **Smart Cargo Storage System – Airport Safety & Security** *(Arduino UNO, RFID, MQ-2)*: Designed an automated cargo safety solution with RFID access control, entry counting, automatic fire detection and suppression, and smart lighting to minimize human error. [GitHub]
- **CleanCare – Laundry Service Management System** *(PHP, MySQL, JavaScript, Chart.js)*: Built a structured, role-based web application managing laundry operations with separate dashboards for Admin, Staff, Delivery, and Customers to handle order tracking, status history, notifications, and CSRF-protected workflows. [GitHub]
- **Patient Care System – Curex Mobile App** *(Figma, Prototyping)*: Designed a healthcare app connecting patients with doctors, pharmacies, and emergency services, including SOS alerts, ambulance booking, an interactive body map, and nearby hospital maps. [GitHub]
- **IRONPULSE – Gym Management System** *(HTML, CSS, JavaScript, PHP, MySQL)*: Developed a responsive gym management platform with Admin and Trainer dashboards, member management, trainer assignments, workout routines, diet planning, and progress tracking. [GitHub]
- **ExpenseMate – Mobile App** *(Figma, Low-Fidelity Prototyping)*: Designed a mobile expense-tracking app for university students and young professionals to log income/expenses and stay on top of their monthly budget. [GitHub]
- **Crimson Cup Bangladesh — Website UI Redesign** *(Figma, Heuristic Evaluation)*: Led a usability-focused redesign including a Nielsen heuristic evaluation, mood board, style guide, and responsive high-fidelity mockups for desktop and mobile. [GitHub]
- **Pawsitive Tails Veterinary Care** *(HTML, CSS, JavaScript)*: Built a frontend demo of a veterinary care platform where pet owners can manage their pets' healthcare in one place. [GitHub]
- **Pocket Arcade -— combined of four mini games** *(HTML5 Canvas, CSS, JavaScript)*: Developed a responsive browser-based arcade featuring six games: Snake, Tic Tac Toe, Memory Match, Whack-a-Hamster, Traffic Dodge, and Bubble Shot. Implemented local high-score storage, light/dark themes, sound effects, and keyboard and touch controls. [GitHub]

Volunteer Experience

IDP Education Ltd August,2026 – August,2026
Volunteer

- Represented Navitas as a volunteer at the IDP Study World Dhaka Roadshow 2026, held at Hotel InterContinental Dhaka on 29 August 2026, assisting students and supporting event operations.
- Volunteer Certificate: [View Certificate]

Certifications

- **AI Workflows and Agents** — GP Academy, 2026 [Certificate]
- **AI+Superpresentations** — GP Academy, 2026 [Certificate]
- **GenAI & Prompt Engineering** — GP Academy, 2026 [Certificate]
- **AI-Pro Aptitude Hacks** — GP Academy, 2026 [Certificate]
- **Build ATS-Winning CV** — GP Academy, 2026 [Certificate]
- **Data Foundations with AI** — GP Academy, 2026 [Certificate]
- **Build Your Future Before Graduation** — GP Academy, 2026 [Certificate]

Achievements

- **Academic:** Received 9 scholarships with 100% tuition waivers (2), 50% waivers (5), and a 25% waiver (2), maintaining tuition-waiver status through all trimesters.
- **BeCrowd:** Solved 150+ coding problems.